

Junzhe Shao

Berkeley, CA | (+1) 858-766-1048 | junzhe_shao@berkeley.edu | [Website](#)

Education

University of California, Berkeley Ph.D. Candidate in Biostatistics	Sep 2023 – 2026 (Expected)
Columbia University M.S. in Biostatistics	Sep 2021 – May 2023
Peking University B.S. in Biological Science; Minor in Physics	Sep 2016 – Jun 2021
University of California, San Diego Visiting Student	Sep 2019 – Sep 2020

Work Experience

Internship, Data Science Research ByteDance (TikTok), San Jose, CA	May 2026 – Present
- Developing cluster-based traffic splitting for recommendation experiments to improve assignment purity and reduce cross-cluster interference. - Designed a Leiden loss approximating the mean squared error (MSE) objective for cluster randomization, improving the purity-precision Pareto frontier on billion-user-scale data. - Developing covariate-adjustment methods for cluster-randomized experiments, drafting technical documentation, and supporting partner teams with methodology design.	
Internship, Product Development Data Science Genentech, South San Francisco, CA	Jun 2025 – May 2026
- Developed a bias-corrected Augmented Inverse Probability Weighting (AIPW) estimator for longitudinal external-control studies, leveraging double machine learning and semiparametric efficiency theory to improve causal effect estimation. - Built reproducible R simulation pipelines to benchmark performance and drafted the methods manuscript. - Applied and validated on real clinical-trial data, delivering study-ready analyses.	

Research Experience

Bias Corrected AIPW Estimator for External Control in Longitudinal RCTs	May 2025-Present
- Developed a bias-corrected augmented inverse probability weighting (AIPW) estimator that leverages external control arms with longitudinal data. - Established bias correction at a rate governed by the number of time points, ensuring valid type I error control compared to naive direct pooling approaches. Provided non-asymptotic guarantees. - Demonstrated higher statistical power compared to RCT-only estimators and other covariates adjustment methods.	
Nonstationary Adaptive A/B Testing with State-space Models	May 2025-Jan 2026
- Proposed a state-space modeling framework for adaptive AB testing in nonstationary environments with evolving treatment effects. - Developed a Neyman-style optimal allocation policy via Kalman filtering and established asymptotic variance bounds for ATE estimator and provide anytime valid confidence interval.	
Regularized Frameworks for Adaptive Enrichment Designs	Jun 2023 – Dec 2025
Proposed a unified adaptive enrichment framework to maximize power for testing average treatment effects.	

- Cast the problem as constrained optimization to unify different objectives.
- Calibrated selection bias and provided valid inference throughout the process.

Enhanced Large Language Model for Predicting HIV Care Disengagement

Dec 2024 – April 2025

- Fine-tuned LLaMA 3.1 with over 4.8 million EMR records from Tanzania to predict disengagement from HIV care, ART non-adherence, and adverse outcomes.
- Demonstrated superior predictive performance compared to supervised ML models and zero-shot LLMs, with strong internal and external validation results.
- Provided clinically interpretable insights; expert evaluation confirmed the model's reasoning was clinically relevant in over 90% of aligned cases.

Causal Inference for Non-Stationary Time Series Data

Nov 2021 – Present

- Proposed a parametric generalization of synthetic control with time-varying, non-permutation-invariant weights.
- Applied the approach to study the impact of a mandatory certificate on COVID-19 vaccine compliance.
- Developing the R package [SSMimpute](#) for state-space-model multiple imputation in non-stationary multivariate time series.

Adaptive Design for Randomized Trials under High-dimensional Setting

Jun 2022 – Present

- Investigated theoretical properties of covariate-balancing methods for randomized clinical trials under many covariates.
- Proved that a balancing criterion can be achieved asymptotically when the number of covariates is a vanishing proportion of the number of patients.

Publications & Manuscripts in Preparation

Junzhe Shao, Xiaoxuan Cai, Linda Valeri. “Synthetic Control as Missing Data: A Unified State-Space Approach”, Preparing for submission.

Junzhe Shao, Jiawen Zhu, Herbert Pang. “Leveraging Longitudinal Data in Designing Hybrid RCT with the Incorporation of Real-World Data”, Submitted to *Biostatistics*, 2026.

Junzhe Shao, Waverley Wei, Jingshen Wang. “Adaptive A/B Testing under Nonstationary Dynamics using State-Space Models”, [International Conference on Artificial Intelligence and Statistics \(AISTATS\)](#), 2026.

Junzhe Shao, Aibo Gong, Juan Shen, Waverley Wei. “A Unified Adaptive Enrichment Design for Power Enhancement”, Major Revision at *Statistica Sinica*, 2025.

Waverly Wei[†], **Junzhe Shao**[†], Rita Qiuran Lyu, Rebecca Hemono, Xinwei Ma et al. “Enhanced language models for predicting and understanding HIV care disengagement: a case study in Tanzania”, [npj Digital Medicine](#), 2026.

Junzhe Shao, Mingzhang Yin, Xiaoxuan Cai, Linda Valeri. “Generalized Synthetic Control Method with State-space Model”, short version accepted at [NeurIPS 2022 Workshop for Causality for realworld impact](#). Oral Presentation at American Causal Inference Conference (ACIC) 2023, Manuscript in preparation.

Alton Barbehenn, Lei Shi, **Junzhe Shao**, et al. “Rapid Biphasic Decay of Intact and Defective HIV DNA Reservoir During Acute Treated HIV Disease”, [Nature Communications](#), 2024.

Mona Alotaibi[†], **Junzhe Shao**[†], Michael W. Pauciulo, et al. “Metabolomic Profiles Differentiate Scleroderma-PAH from Idiopathic PAH and Correspond with worsened Functional Capacity”, [Chest](#), 2022.

[†] Co-first authors.

Skills Summary

Programming: R, Python, SQL

Software/Tools: Spark, Ray, Hadoop, Hive, $\LaTeX$, Git, Bash, Slurm

Languages: Mandarin (Native), English (Proficient)